How to Estimate RPS

DAU, reads vs writes, fan-out and peak

Watch the 3-min video

In one line: RPS = DAU × actions per user per day ÷ 86,400 (≈ 10^5). Then split reads from writes, multiply by the calls each action makes, and apply a peak factor. Size for the peak.

1 The question behind the number

Your interviewer says: "a hundred million daily active users, and each one reads their feed ten times a day. How many requests per second must the backend handle?" If you can turn that sentence into a number in thirty seconds, you will know whether you need **one server or a thousand**.

2 The core formula

```
RPS = DAU × actions per user per day ÷ 86,400
```

A day has 86,400 seconds. For quick math, round it to **100,000**. Your answer comes out about 14% low, which is fine for an estimate: you only need the right order of magnitude.

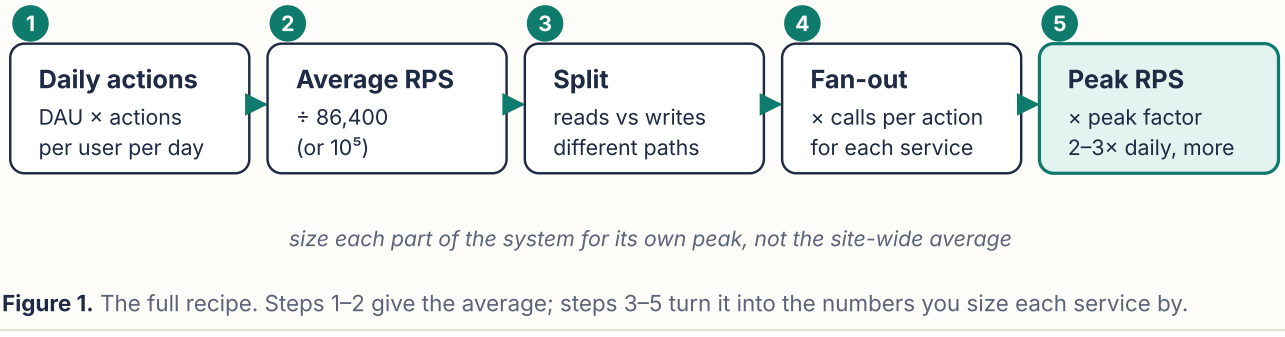


Figure 1. The full recipe. Steps 1–2 give the average; steps 3–5 turn it into the numbers you size each service by.

WORKED EXAMPLE: THE FEED

100 M users × 10 reads = **1 billion reads a day**. Divide by 86,400 → about **11,600 reads per second** on average. With the rounded shortcut, $10^9 \div 10^5$ = **10,000**. Either answer is the right order of magnitude.

3 Always split reads and writes

Reads and writes hit **different parts of the system**. Reads go to caches and read replicas, which scale out easily. Writes go to the primary database and to queues, which are harder to scale. So a single "total RPS" hides the number that matters for each part.

If those same 100 M users each post once a day, that is 100 M writes a day, about **1,200 writes per second**, against about **11,600 reads per second**.

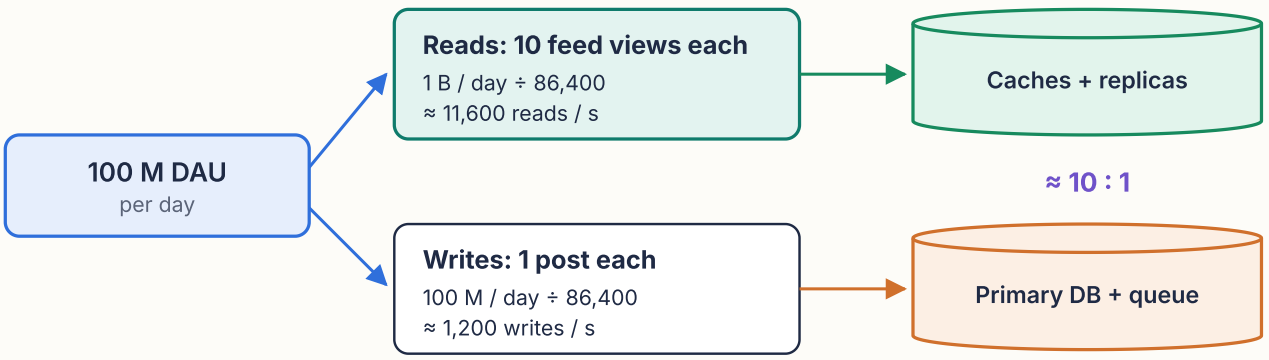


Figure 2. The same users produce two very different loads. Size the read path and the write path separately.

4 One action is rarely one request: fan-out

Opening a feed might call the feed service, the user service for avatars, the like counts and the ads service. That is a **fan-out** of four or more. So, for each service you size, multiply by the number of calls it receives per user action. **Internal traffic is often several times larger than external traffic.**

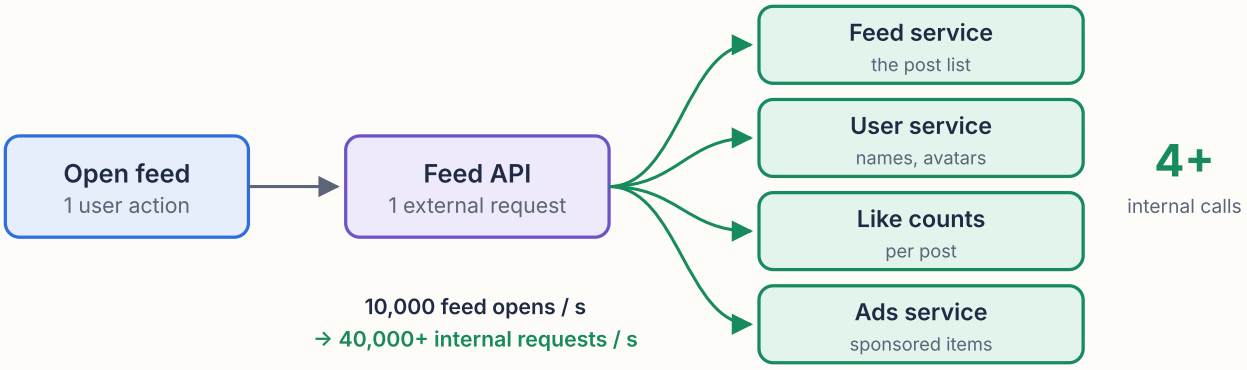


Figure 3. One tap becomes one external request and at least four internal ones. The user service sees every feed open, not just profile views.

5 Then apply the peak factor

Everything so far is the **average**. Real traffic follows the day, with evening peaks, and has sudden spikes. Multiply the average by a **peak factor**: often **2–3×** for daily patterns, and much more for events. Size the system for the peak, not the average. The next lesson covers this in detail.

WORKED EXAMPLE: A CHAT APP

- **Input:** 50 M DAU, 20 messages each per day → **1 billion messages a day**.
- **Average writes:** $10^9 \div 10^5 \approx 10{,}000$/s; more precisely $10^9 \div 86{,}400 \approx$ **11,600 messages/s**.
- **Reads:** each message is delivered to at least one recipient, so reads are at least as high: ≥ 11,600/s, and more for group chats.
- **Peak:** at a 3× peak factor, plan for about **35,000 writes/s**.

COMMON MISTAKES

- Giving one total RPS and never splitting reads from writes.
- Forgetting fan-out, so internal services are sized for a quarter of their real load.
- Sizing for the average and ignoring the peak.
- Spending a minute on exact long division when 10^5 gets the order of magnitude in seconds.

6 Key terms

RPS / QPS: requests (queries) per second

DAU: daily active users

Actions per user: how often each user reads or writes per day

Read/write split: separate estimates for each path

Fan-out: internal calls made per user action

Peak factor: peak ÷ average, often 2–3× daily

7 Interview questions

Q1. How do you estimate requests per second? BASIC

DAU × actions per user per day ÷ 86,400 (or 10^5 for quick math). Then split reads from writes, multiply by calls per action, and apply a peak factor.

Q2. Why split reads from writes? BASIC

They hit different parts of the system. Reads go to caches and replicas, writes to the primary database and queues. 11,600 reads/s and 1,200 writes/s call for very different designs.

Q3. How wrong is it to divide by 100,000 instead of 86,400? BASIC

About 14% low (86,400 ÷ 100,000 = 0.864). For an estimate that is fine: it never changes the order of magnitude.

Q4. 50 M DAU, each sends 20 messages a day. Estimate RPS. SCENARIO

1 B messages/day ÷ 86,400 ≈ 11,600/s on average (≈ 10,000 with 10^5). Reads are at least as high. At a 3× peak factor, plan for about 35,000 writes/s.

Q5. 100 M DAU open the feed 10 times a day, and each open calls 4 services. How much traffic does the user service get? SCENARIO

Feed opens: 1 B/day ≈ 11,600/s on average. If every open calls the user service once, it sees ~11,600/s too; the four services together receive ~46,000 internal requests/s. At 3× peak, the user service needs ~35,000/s.

Q6. An estimate gives 500 RPS at peak. What does that tell you? SCENARIO

A couple of ordinary servers (plus one for redundancy) and a single database can handle it. You don't need sharding or a complex cache layer yet.

Q7. Why is internal traffic often larger than external traffic? TRICKY

Fan-out: one user action calls several services, and those may call others. A page that calls 4 services, each making 2 database queries, turns 1 external request into 12 internal operations.

Q8. Your DAU is 10 M but MAU is 100 M. Which do you use for RPS? TRICKY

DAU: it drives daily traffic. MAU matters for storage (accounts, profiles) and for growth. Using MAU would overestimate daily traffic about 10×.

Next: SD-D-004 · Peak vs Average Traffic backendroadmap.com